Static Timing Analysis

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Administration

Total 20H = 12H Lectures + 8H Laboratory works

Course mailing list: send e-mail to vazgenm@synopsys.com

Grading

Grading will be assigned on:

- Lectures (70 scores)
- Laboratory works (30 scores)

Final exam

Course Overview

1. STA Concepts
   - 2 lectures
2. Delay Modeling
   - 2 lectures
3. Interconnect Parasitics
   - 2 lectures
4. Delay Calculation
   - 2 lectures
5. Configuring the STA Environment
   - 3 lectures
6. Timing Checks
   - 1 lecture
7. Crosstalk and Noise
   - 2 lectures
8. Statistical Static Timing Analysis
   - 2 lectures

References

1. V. Taraate. Digital Design from the VLSI Perspective: Concepts for VLSI Beginners. Springer; 1st ed. 2025 edition ; 2025
2. R. Sharma. Characterization and Modeling of Digital Circuits. Independently published ;2018
3. V. Champac, J. G.Gervacio. Timing Performance of Nanometer Digital Circuits Under Process Variations. pringer; 1st ed. 2018 edition; 2018
4. S. Gangadharan., S. Churiwala. Constraining Designs for Synthesis and Timing Analysis: A Practical Guide to Synopsys Design Constraints (SDC). Springer; 2015

Thank You